

The Icosahedral Origin of $\theta = 2/9$: the Second Witness

How the C_3 -in- A_5 doubling produces the Koide angle parameter-free and exactly —
and why geometry and phenomenology coincide here

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Correction notes

Corrections from the calculation check (30 Sept. 2026). The following places are corrected or annotated in the text; each carries a dated note with the previous value:

- Section 1: leak $(7 - 3\varphi)/9$ against δ^* : "at the tolerance 0.135 %, $\sim 1.3\sigma$ " $\rightarrow$ just above (0.183 %, $\sim 1.4\sigma$).
- Section 1 (1 Oct. 2026): tolerance from Doc. 291 with the local slope 0.052 % instead of 0.135 %; the gap is then $\approx 3.5\sigma$ (annotated).

.1 Question: a second, independent route to $\theta = 2/9$

The Koide angle $\theta = 2/9$ is empirically confirmed to seven digits and understood in FFGFT as the rational address of a value fixed through a reference point (Dok. 292), not as a pole-mass exactness claim (P42). Dok. 290/291 established the *channel* in which θ lives: not the symmetric magnitude functional (where $2/9$ is counter-blind), but the phase channel, with $\chi = \pi/2$ as the skew-adjoint selector and δ as the still-open second harmonic. The *value* of θ remained a found, not a derived, number.

This document supplies a second, entirely independent route. It comes neither from the masses nor from the Koide parametrisation, but purely from the *geometry* of the doubling $6 = 3 \oplus 3'$ (Dok. 285): the embedding of FFGFT's own C_3 structure into the icosahedral ambient group A_5 . This embedding produces $\theta = 2/9$ parameter-free, forward and exactly – with no $2/9$ as input. That a second, methodologically foreign route hits the *same* rational value is the actual statement: not a translation of one into the other, but a convergence in the sense of the established FFGFT logic, in which α , the lepton masses and Koide act as *verifications* of distinct routes pointing at the same constants. The computations are in `ikosaeder_theta_delta.py` (numpy-only, seed 20780458).

.2 The C_3 -in- A_5 embedding and the full redistribution

The three lepton modes are the Fourier eigenmodes of the Z_3 circulant on the $\mathbb{C}^3$ fibre (Dok. 282/288): the trivial mode $v_0 = \frac{1}{\sqrt{3}}(1, 1, 1)$ and the two non-trivial, Galois-conjugate modes $v_{1,2} = \frac{1}{\sqrt{3}}(1, \omega^{\pm 1}, \omega^{\pm 2})$ with $\omega = e^{2\pi i/3}$. The electron is a non-trivial mode; which of the two is decided solely by the sign of the angle, not by the value computed here.

The doubling places C_3 inside the icosahedral group A_5 ; the five-fold generator is a rotation R_5 by 72° about an icosahedral 5-fold axis (standard embedding: axis $(0, 1, \varphi)$ with $\varphi = \frac{1+\sqrt{5}}{2}$). Since the 3-fold and 5-fold axes point in different directions, R_5 mixes the C_3 modes. The *full* redistribution of the electron mode $v_e = v_1$ has weights $p_j = |\langle v_j | R_5 v_e \rangle|^2$:

$$\underbrace{p_0 = \frac{2}{9}}_{\text{trivial } v_0}, \quad \underbrace{p_2 = \frac{5-3\varphi}{9}}_{\text{2nd harm. } v_2}, \quad \underbrace{p_1 = \frac{2+3\varphi}{9}}_{\text{stays } v_e}, \quad p_0 + p_1 + p_2 = 1.$$

All three weights carry the denominator $9 = 3^2$ – the same Z_3 structure in which $2/9$ sits as an orbit address. Confirmed numerically (40 digits, mpmath): $p_0 = 0.2222\dots = 2/9$ exactly, difference $< 10^{-30}$.

.3 The decisive finding: $\theta = 2/9$ as the trivial share

The share that, under the icosahedral 5-fold, passes into the *trivial* mode v_0 is

$$p_0 = |\langle v_0 | R_5 v_e \rangle|^2 = \frac{2}{9}$$

exactly and parameter-free. No mass ratio, no Koide parametrisation and no $2/9$ enters the value; it follows from the C_3 modes and the icosahedral rotation alone. The associated *total* leak is $p_0 + p_2 = (7 - 3\varphi)/9 = 0.23843$, the second, parameter-free candidate for the amplitude δ of the second harmonic; it agrees with the $\delta^* = 0.23887$ back-computed from $\theta = 2/9$ to 0.18 % (just above the τ -limited tolerance of 0.135 %, $\sim 1.4\sigma$) (*Corrected on 30 Sept. 2026: previously "just at the tolerance ... $\sim 1.3\sigma$ "* – $(7 - 3\varphi)/9 = 0.238433$ against $\delta^* = 0.238871$ gives -0.183 %, and $0.183/0.135 = 1.36$ lies above the 1σ tolerance.) (Note of 1 Oct. 2026: The tolerance 0.135 % comes from Doc. 291, where it rests on the median slope; with the local slope it shrinks to 0.052 % (note in Doc. 291). The deviation -0.183 % then corresponds to $\approx 3.5\sigma$; the leak $(7 - 3\varphi)/9$ is not compatible with δ^* (Doc. 190, R139).). Value and angle thus stem from the same 5-fold redistribution: θ is the trivial, δ the total share. The earlier separate treatment as single harmonics (first for θ , second for δ) was the too-narrow view; the full redistribution carries both.

.4 Robustness: the specificity of the icosahedral geometry

The value $2/9$ is not a generic property of any rotation, but of the icosahedral one:

- **Axis specificity.** 200 random rotation axes at 72° give the v_0 leak in the range $[0.036; 0.452]$; not one hits $2/9$. Only the icosahedral 5-fold axes do.
- **Angle and order specificity.** The angle scan shows $2/9$ exactly at 72° ; the n -fold series yields $2/9$ only at $n = 5$ ($2/5$ at $n = 3$, $4/9$ at 144°). The rational cleanliness occurs at the distinguished angles.
- **Mode independence.** Both non-trivial modes v_1, v_2 give $p_0 = 2/9$; the trivial v_0 would give $4/9$. The result does not depend on which conjugate mode is read as the electron.
- **Axis choice (discrete, P3).** Of the six 5-fold axes, three ($+\varphi$ branch, 3-5 axis angle 37.38°) give $2/9$, three ($-\varphi$ branch, 79.19°) give $4/9$. The physical embedding $C_3 < A_5$ selects the nearest 5-fold, the $+\varphi$ branch – hence $2/9$. It is a discrete two-way choice, not a free parameter.

.5 Two witnesses, one value

The θ question thus has two mutually independent sources that coincide exactly:

Route	Origin	Value
Koide (phenomenological)	from the measured lepton masses; no geometry	$\theta = 2/9$
C_3 -in- A_5 (geometric)	from the 5-fold redistribution of the modes; no masses, no $2/9$ input	$p_0 = 2/9$

The two routes know nothing of each other – one knows only masses, the other only geometry – and yet hit *exactly* the same rational number. This is a convergence, not a

translation deficit: it is not required that the geometric route translate back into the Koide cosine parametrisation. Just as a derived value need not be identical to its measurement prescription to confirm it, the parameter-free geometric prediction $2/9$ need not take the shape of the phenomenological fit parameter to agree with it. That the geometric value does *not* vary with θ is here the hallmark of a prediction, not a defect: the geometry has no free parameter and names a fixed value.

.6 $\theta = 2/9$ as a translation invariant of two models

The “second witness” must be stated more precisely so as not to be misread as an ontological claim. It is *not* about a geometric truth beating a phenomenological one, nor about the torus “really” being the Hilbert space or vice versa. Both – the point/differential geometry of the T^4 and the spectral geometry of the Hilbert space $L^2(T^4) \otimes \mathbb{C}^3$ – are *mathematical models* that are demonstrably and losslessly translatable into one another (Dok. 230/282): the Z_3 fibres, the C_3 Fourier modes, the A_5 embedding are the same structure in both languages, merely represented differently.

The actual finding is therefore methodological, not ontological: $\theta = 2/9$ is an *invariant of the translation*. It appears in the spectral representation (as the redistribution weight p_0) just as in the phenomenological Koide address, because it belongs to the shared structure and not to a single representation. This is precisely what makes it robust: the constant falls out of the structure irrespective of which of the translatable languages one computes in. This lies on the established line in which $\hbar$ and c are conversion factors and not ontological primitives: what counts is what is invariant under the conversion, not the choice of one side as “true”. On the question of which of the two models is reality, nothing is asserted here deliberately – it is not needed, and the computation does not supply it. The value of the result lies precisely in its *model independence*.

.7 Scale invariance of the pattern versus occupation

No scale enters the geometric value $p_0 = 2/9$ or the ratios: the C_3 -in- A_5 geometry knows neither mass nor energy, it yields pure numbers – the angle and the denominator-9 weights. The *pattern layer* – angle plus ratios – is therefore a property of the topology, not of a particular energy range. It holds wherever the Z_3/A_5 doubling occurs; the lepton sector is *one* realisation, namely the one in which the scale E_0 (Dok. 292) places the absolute position exactly on the measured e, μ, τ .

This is to be strictly separated from the *occupation layer*: whether states actually sit at a given scale. The geometry says “*if* a Z_3 triplet is realised here, then with this θ and these ratios” – it does not say “here is one”. At another scale the same circulant pattern stands without occupied states having to lie there. This is the same logic as the tower $D(4 + 3n)$ and the emptiness theorem (Dok. 285): the pattern is present everywhere, the occupation is a separate, scale-bound and dynamics-dependent question; an empty window carries the same pattern as an occupied one, only without particles in it. From the scale invariance of the pattern there follows therefore *no* prediction of new particles at other scales – the universality concerns the structure, not the occupation.

.8 Status and open edge

What is secured: The icosahedral doubling structure produces $\theta = 2/9$ parameter-free, forward, exactly and icosahedron-specifically, in independent agreement with the phenomenological Koide value; the amplitude $\delta = (7 - 3\varphi)/9$ stems from the same redistribution. Thereby the long-open question – whether the T^4/Z_3 structure forces $\theta = 2/9$ – has, for the first time, a concrete, exact, geometric answer within the already established A_5 doubling (Dok. 285).

What remains open as a derivation step (and explicitly not as an objection to the convergence): the explicit mechanism linking the geometric redistribution weights $(2/9, (5 - 3\varphi)/9, (2 + 3\varphi)/9)$ to the physical mass ratios. This bridge need not run through Koide but can be led directly through the masses or the mode structure; it is the subject of further work. The operational role of the 5-fold in the phase channel is consistent with Dok. 291: R_5 sets a phase $\pi/2$, the value Dok. 291 associates with χ .